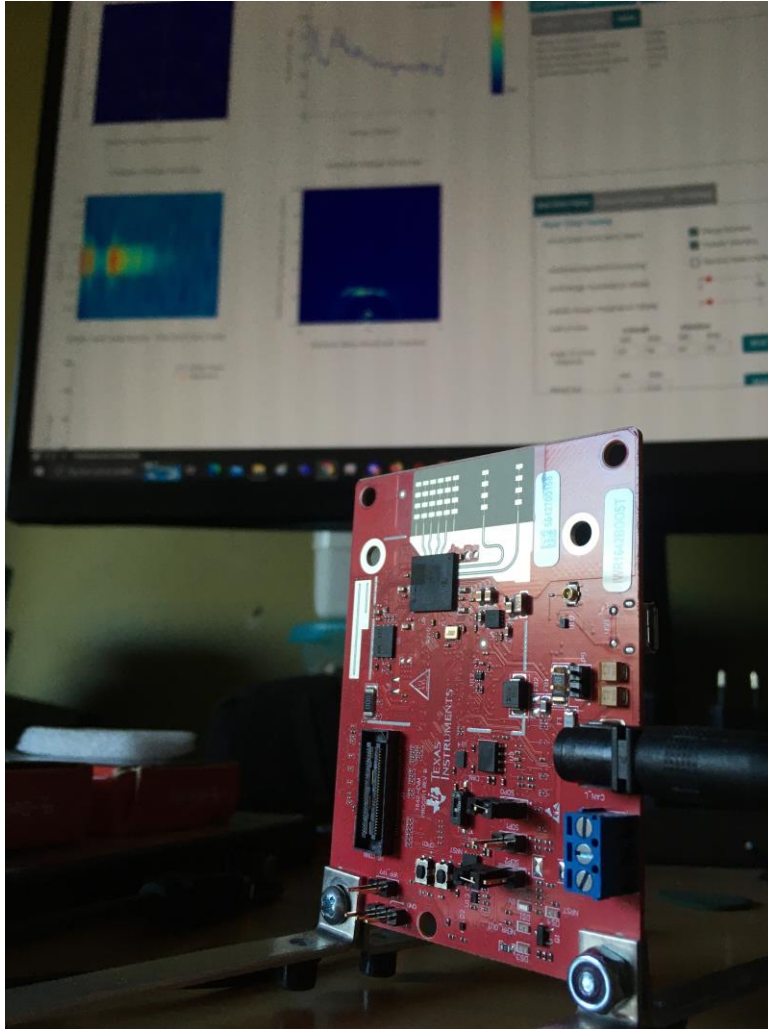


# TI IWR1642boost Object Detection



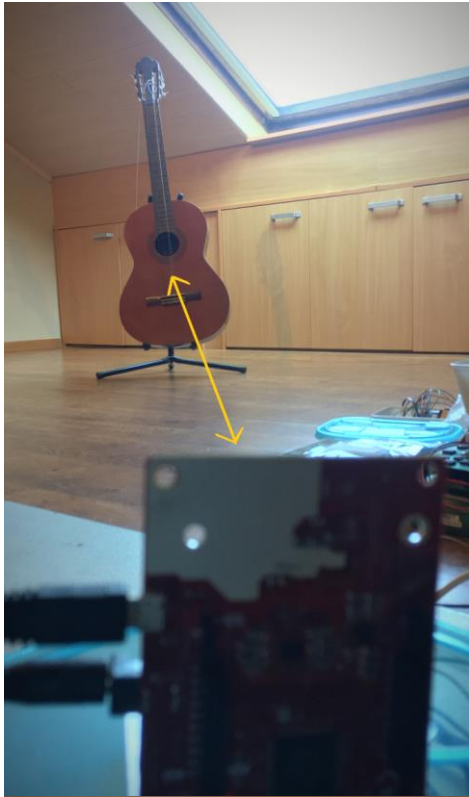
## Setup material:

- TI IWR1642boost module
- TI IWR1642 Out-Of-The-Box Object Detection Demo software
- USB connected laptop running the TI Demo Visualizer in Chrome
- Target object: Guitar

## Goal:

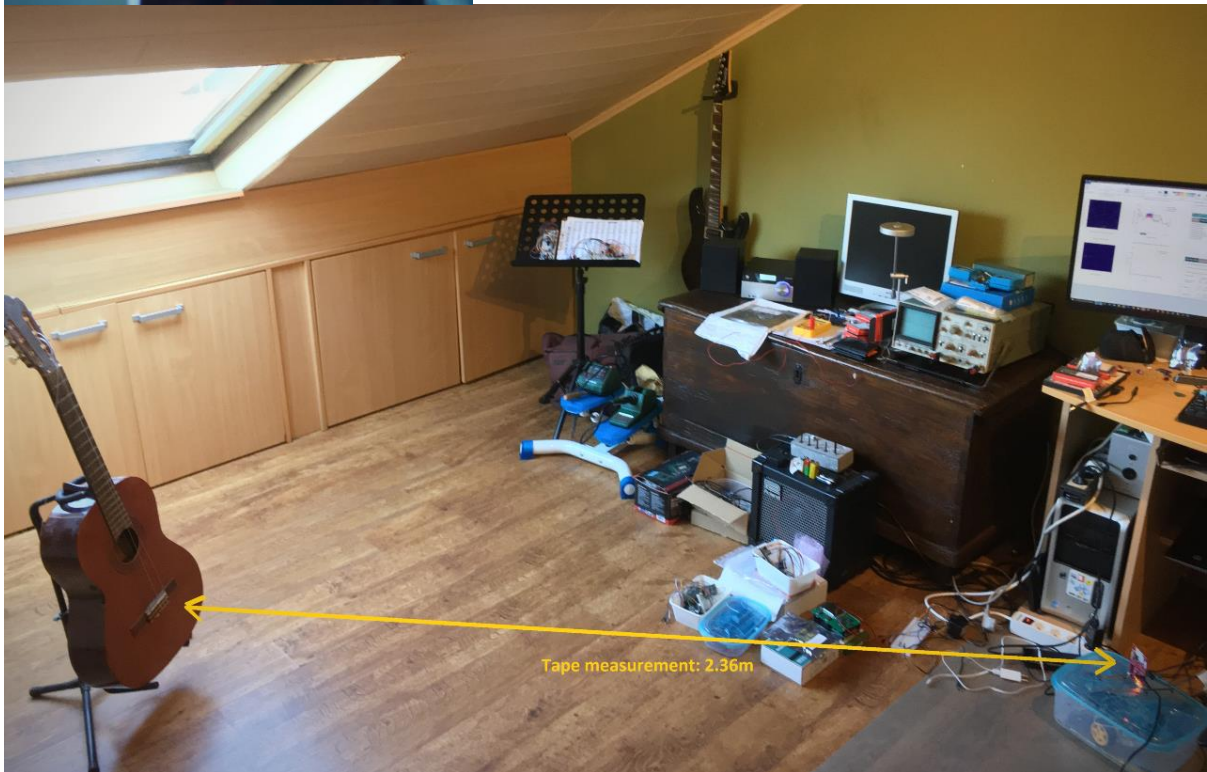
- Test the complete TI IWR1642 toolchain
- Get the very first measurement data out of the device
- Play around with different configurations using the mmWave Demo visualizer to gain a first hands-on experience with the TI IWR1642 basic functionality

## TI IWR1642 Measuring Setup:



In this setup, a guitar is placed at a certain distance from the mmWave sensor. The guitar is a fixed object and thus the measurement will happen on a static object with no velocity. One of the reasons a guitar is chosen as the object is because of its material. The coated wood has good reflectivity and thus increases the Radar Cross Section (RCS) which will result in reliable detection for this test setup.

The distance between the guitar and the TI IWR1642 boost is measured with a tape measure. This is not meant to be accurate and is only used for a quick and dirty comparison with the TI IWR1642 measurements. Also, the TI IWR1642 is not calibrated in this test so the measurements will also be inaccurate.



# TI IWR1642 Measuring Results – mmWave Demo Visualizer:

## Main Configuration of the TI IWR1642

**mmWave Demo Visualizer** Options Help

### Setup Details

|                                    |                 |
|------------------------------------|-----------------|
| Platform                           | xWR16xx         |
| SDK version (*)                    | 3.6             |
| Antenna Config (Azimuth Res - deg) | 4Rx,2Tx(15 deg) |

### RCS

|                                      |         |
|--------------------------------------|---------|
| Desired Radar Cross Section (sq. m)  | 0.5     |
| Maximum Range for desired RCS (m)    | 39.641  |
| RCS at Max Unambiguous Range (sq. m) | 0.00134 |

### Desirable Configuration

|                               |               |
|-------------------------------|---------------|
| Best Range Resolution         |               |
| Frequency Band (GHz)          | 77-81         |
| Calibration Data Save/Restore | None 0x1F0000 |

### Scene Selection

|                                  |       |
|----------------------------------|-------|
| Frame Rate (fps)                 | 10    |
| Range Resolution (m)             | 0.044 |
| Maximum Unambiguous Range (m)    | 9.02  |
| Maximum Radial Velocity (m/s)    | 1     |
| Radial Velocity Resolution (m/s) | 0.13  |

### Plot Selection

|                                                   |                                                 |
|---------------------------------------------------|-------------------------------------------------|
| <input checked="" type="checkbox"/> Scatter Plot  | <input type="checkbox"/> Range Azimuth Heat Map |
| <input checked="" type="checkbox"/> Range Profile | <input type="checkbox"/> Range Doppler Heat Map |
| <input type="checkbox"/> Noise Profile            | <input checked="" type="checkbox"/> Statistics  |

SEND CONFIG TO MMWAVE DEVICE

SAVE CONFIG TO PC

RESET SELECTION

### Console Messages

```
mmwDemo:/>lvsStreamCfg -1 0 0 0
Done

mmwDemo:/>compRangeBiasAndRxChanPhase 0.0 1 0 1 0 1 0 1 0 1 0 1
0 1 0
Done

mmwDemo:/>measureRangeBiasAndRxChanPhase 0 1.5 0.2
Done

mmwDemo:/>CQRxSatMonitor 0 3 5 121 0
Done

mmwDemo:/>CQSigIntMonitor 0 127 4
Done

mmwDemo:/>analogMonitor 0 0
Done

mmwDemo:/>aoaFovCfg -1 -90 90 -90 90
Done

mmwDemo:/>cfarFovCfg -1 0 0 8.92
Done

mmwDemo:/>cfarFovCfg -1 1 -1 1.00
Done

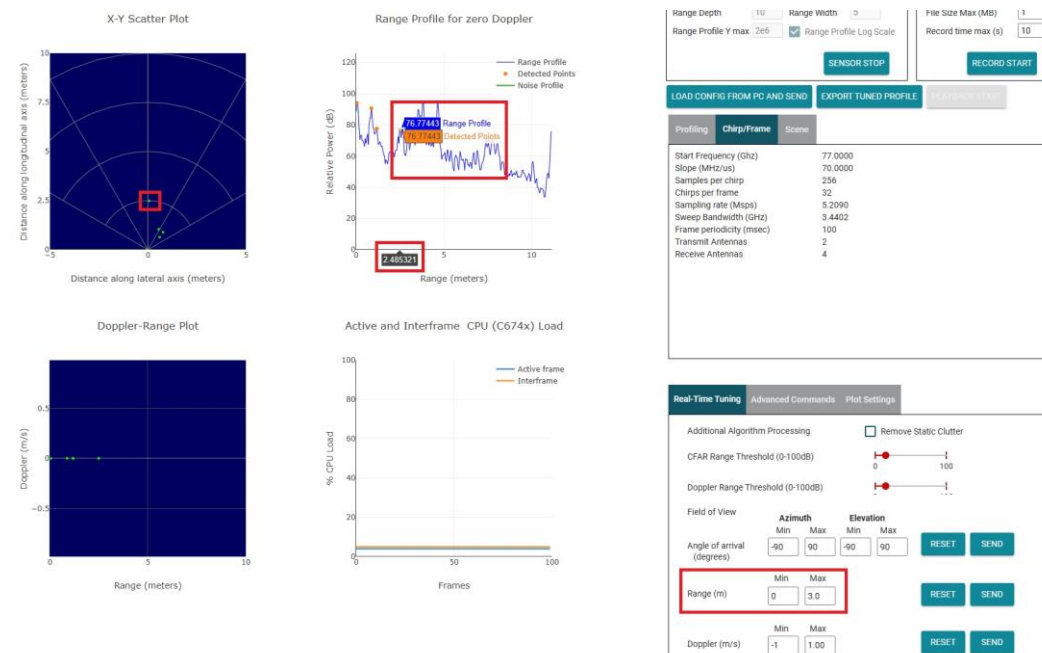
mmwDemo:/>calibData 0 0 0
Done

mmwDemo:/>sensorStart
Debug: Init Calibration Status = 0x17fe
Done
```

CLEAR CONSOLE

## Processed Measurements:

Note: Max range is limited to 3m (Chosen based on the test room dimensions)



## Results:

Tape measured distance: 2.36m

TI IWR1642 measured distance: 2.485m

## Conclusion:

The measured distances only deviate 12.5cm.

These measurements happened without any form of calibration or compensation, so these results will not be accurate but will give a decent approximation.

As the main goal of this demo project is to test the TI IWR1642 toolchain and demonstrate some of the functionality of the IWR1642, it can be concluded that this Demo project succeed in its intention.